



Brain Tumor Spectral Clustering via Feature Extraction



Department of Mathematics, National Taiwan Normal University, Taiwan

Chun-Hao, Wen (文君豪)

Advisor : Prof. Shih-Feng, Shieh (謝世峰)

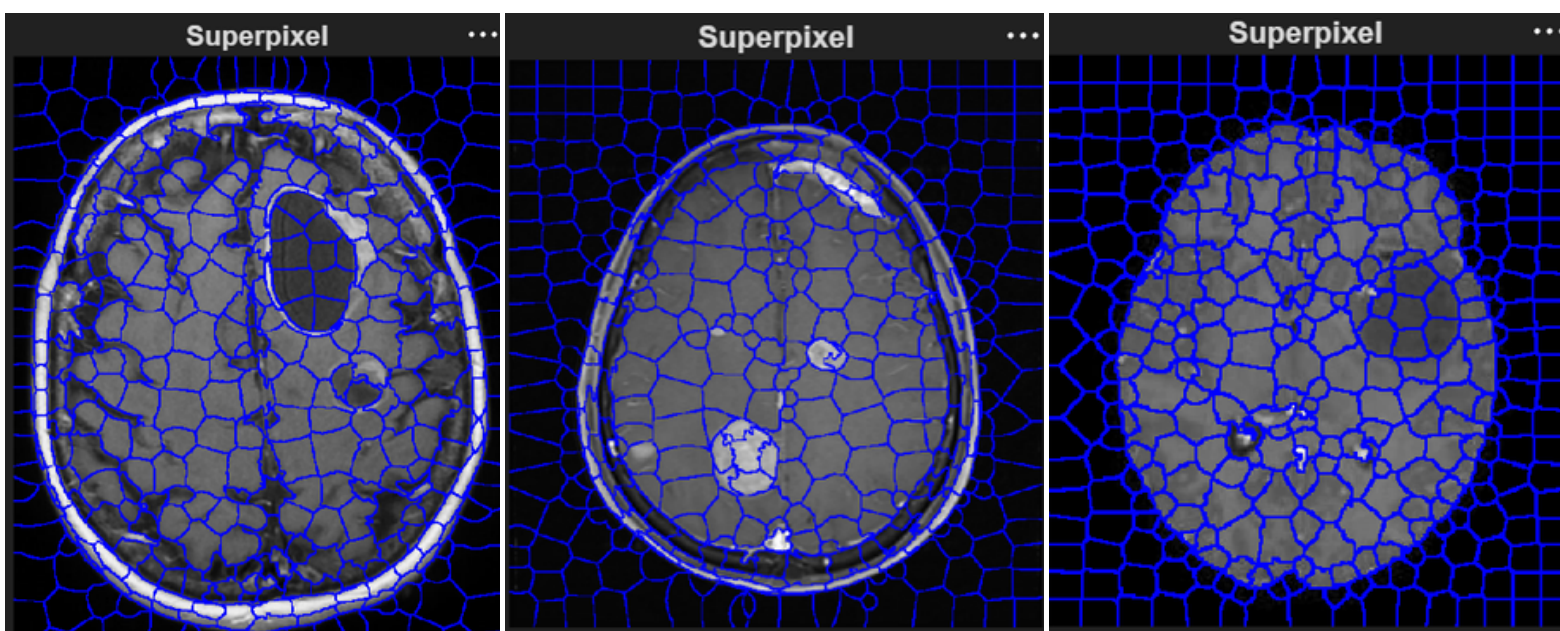
1. Motivation

To realize AI's clinical value, this research focuses on automated brain tumor segmentation to address core challenges:

- **Limitations of Manual Methods:** Manual MRI annotation is time-consuming and error-prone due to blurred boundaries and intensity overlaps with normal tissue, elevating surgical risks.
- **AI-Driven Breakthroughs:** Utilizing objective mathematical computations, our approach effectively overcomes intensity overlaps, rapidly and accurately isolating complex tumor regions.
- **Clinical Impact:** This algorithm delivers a reliable preoperative blueprint, alleviating manual interpretation burdens and significantly enhancing surgical safety and efficiency.

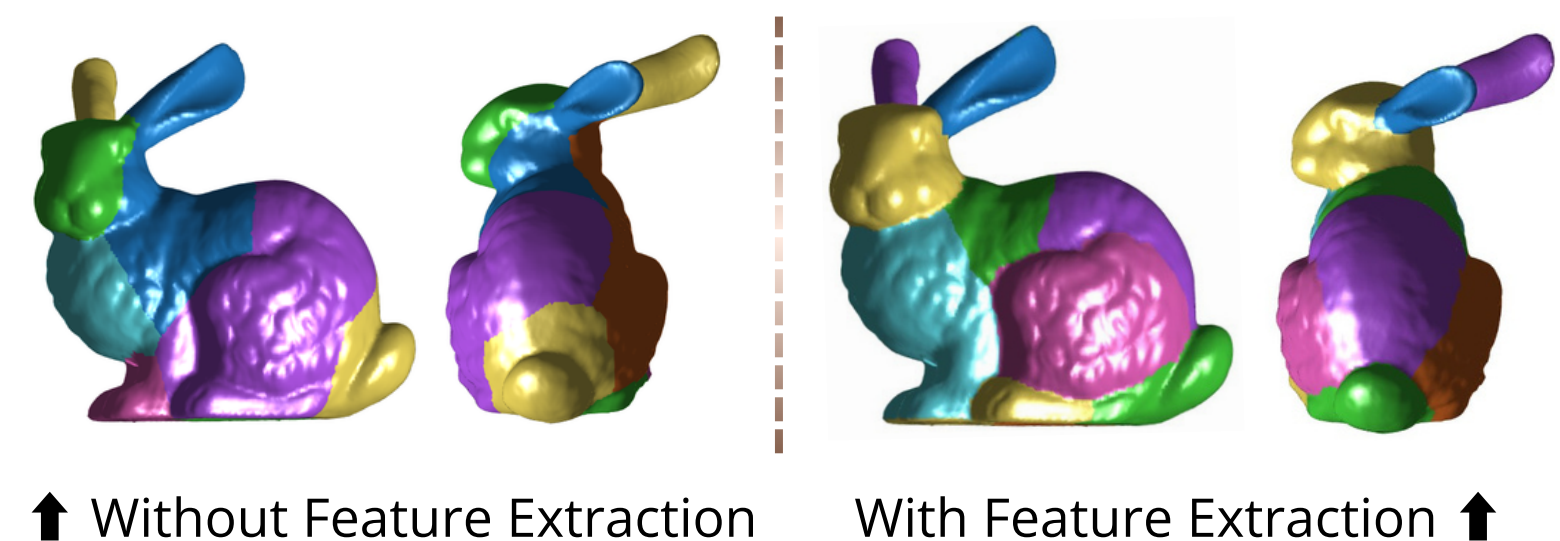
3. Superpixel Algorithm

- **Overcoming Computation Bottlenecks:** Segmenting high-resolution MRIs into superpixels replaces costly pixel-level operations, significantly reducing the dimensionality for the Graph Laplacian matrix (L_{sym}).
- **Homogenization of Local Features:** Grouping adjacent tissues with similar intensities effectively filters out noise while perfectly preserving the original boundaries of the tumor and brain structures.



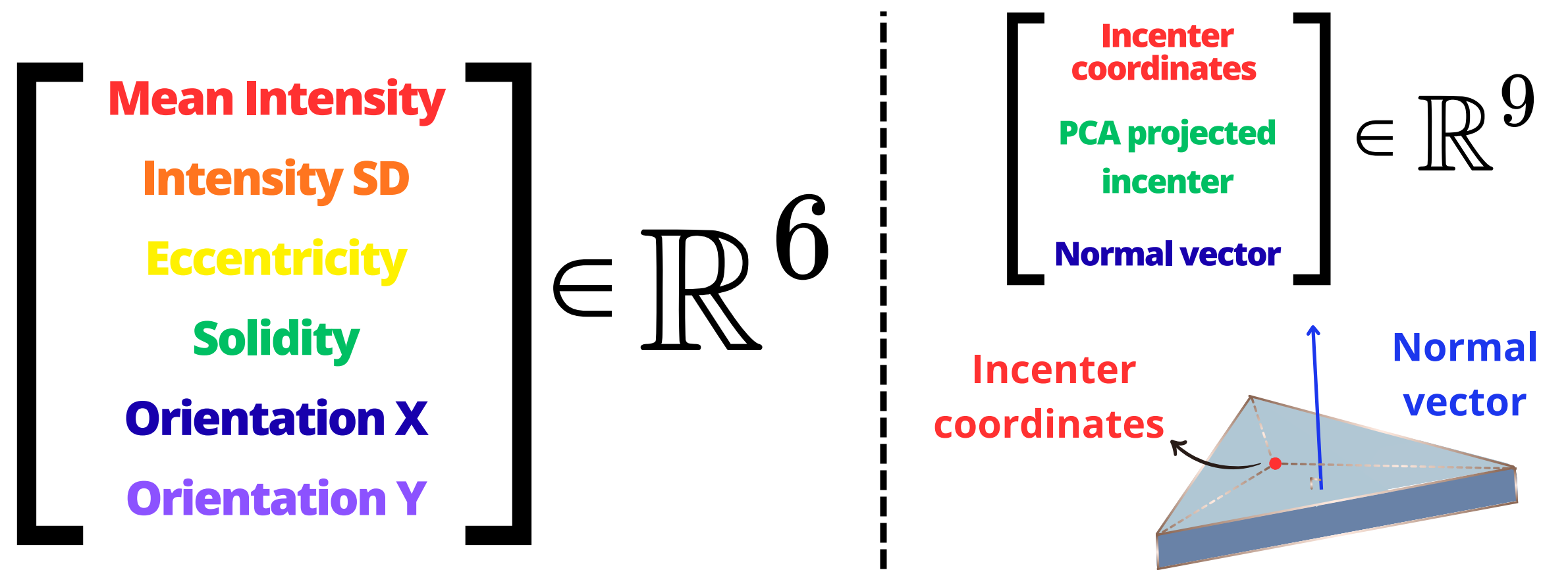
5. Conclusion

- **Clustering Efficiency:** Feature engineering effectively eliminates structural interference, reducing the clusters required for precise segmentation from $k = 7$ down to only $k = 2$ (tumor vs. non-tumor).
- **Robustness via Simplified Logic:** Widened feature margins eliminate the need for multi-criteria evaluation. Relying solely on circularity—completely independent of image intensity—enables the model to robustly adapt to diverse inputs like varying contrast agents.
- **Visualization: Stanford Bunny Part Clustering**



2. Feature Extration Engineering

- **Multi-dimensional Feature Construction (6D Feature Vector):** A dedicated 6D feature vector is constructed for each superpixel block. Distinct from conventional pixel-level analysis, this study comprehensively extracts superpixel features, including regional average intensity, intensity standard deviation, and basic spatial coordinates. This approach provides a robust physical foundation for the subsequent mapping into the spectral space.
- **Weighted Similarity Matrix Calibration:** Based on the optimized feature space mentioned above, feature weights are recalibrated to grant absolute dominance to intensity and internal variance. This calibration successfully guides the subsequent Graph Laplacian matrix (L_{sym}) to accurately reflect the intrinsic textural differences between the tumor and normal brain tissues.



4. Mathematical Model of Spectral Clustering

- **Degree Matrix & Graph Construction:** Treating the feature-extracted superpixels as graph nodes, we define a diagonal Degree Matrix D , where each diagonal element represents the sum of edge weights: $d_i = \sum_{j=1}^n w_{ij}$.
- **Laplacian Matrix Formulation:** We construct the symmetric and positive semidefinite Graph Laplacian Matrix $L = D - W$. This matrix accurately captures the textural boundary differences between the brain tumor and normal brain tissues.
- **Spectral Space Optimization (Main Task):** The core of the clustering algorithm is formulated as an eigenvalue minimization problem. The objective is to find an indicator vector v that minimizes the following function:

$$\min v^T L v = \frac{1}{2} \sum w_{i,j} (v_i - v_j)^2$$

This mathematical constraint ensures that highly similar blocks (with large weight $w_{i,j}$) are forced to have closely aligned vectors ($v_i \approx v_j$), effectively and accurately isolating complex tumor regions in the lower-dimensional spectral space.

- **Tumor Identification:** Initially, tumor identification relied on a complex 6:3:1 weighted scoring system evaluating circularity, intensity, and minor factors. The core geometric metric used is:

$$Circularity = 4\pi \times Area / Perimeter^2$$

However, since feature extraction effectively widens cluster margins, we can now accurately isolate tumors relying solely on this circularity metric.

